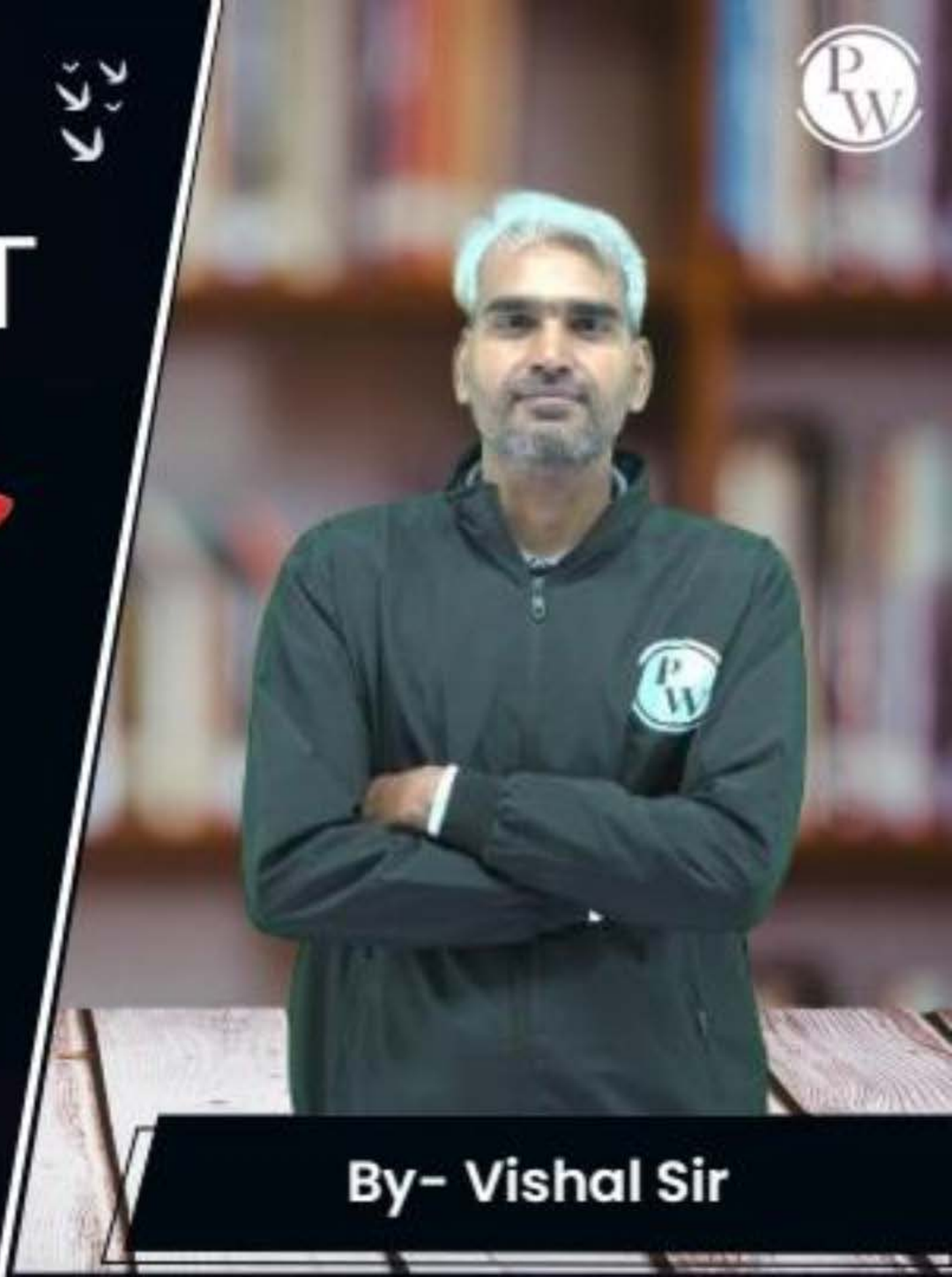
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 10



By- Vishal Sir

Recap of Previous Lecture



Topic Partial Order Relation and Partially Ordered Set
(Ref. + Anti-symm. + Transitive)

Topic Total Order Relation and Totally Ordered Set
{Every pair of elements of set must be Comparable} w.r.t. given relation

Topic Least Upper Bound & Greatest Lower Bound

Slide

Topics to be Covered



- Topic: Minimal / Maximal elements in a POSET
- Topic: Minimum(Least) element of POSET
- Topic: Maximum(Greatest) element of POSET
- Topic: Lattice
- Topic: Types of lattice

Slide



Topic : Partial Order Relation

A relation R on set A is said to be partial order relation if and only if relation R is

① Reflexive
& ② Anti-symmetric
& ③ Transitive

Slide

eg: Let $A = \{1, 2, 3\}$

$R_1 = \Delta_A = \{(1,1), (2,2), (3,3)\}$

- Reflexive ✓
- Anti-symmetric ✓
- Transitive ✓

$\Rightarrow$ ∴ Partial order relation

Diagonal relation on set A is the only relation which is equivalence relation as well as Partial order relation

$R_2 = \{(1,1), (2,2), (3,3), (1,2), (2,3), (1,3)\}$

- Reflexive ✓
- Anti-symmetric ✓
- Transitive ✓

∴ Partial order Relation



Topic : Partially Ordered Set / (P.O. SET)

Let R be a Partial order relation on set A.
then set A along with this Partial order relation R is called Partially ordered set.

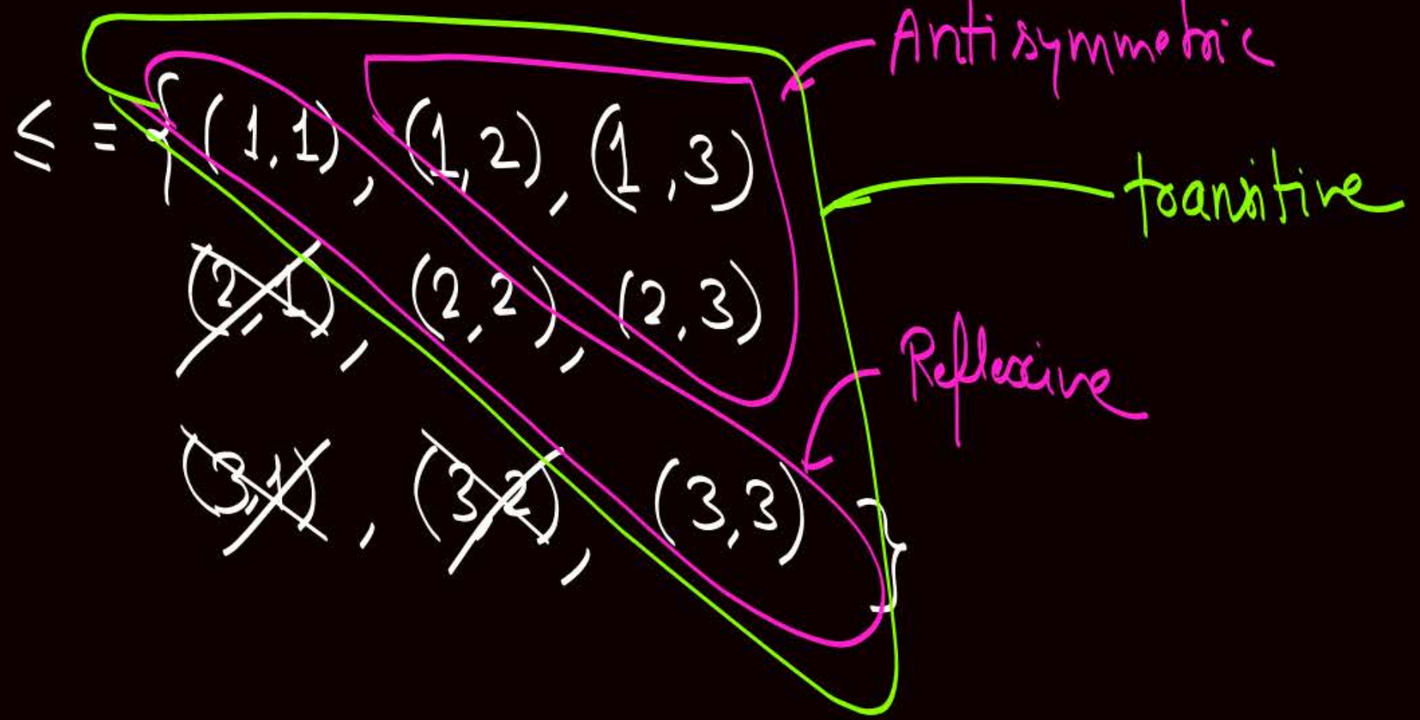
It is denoted by (A, R)

Must be a Partial order relation

Slide

eg: Let 'A' is any set of real numbers, then $(A, \leq)$ is a POSET.

- Let $A = \{1, 2, 3\}$



eg: let A is any set of non-zero positive integers then $(A, \div)$ is a POSET

divides

{ '$\div$' is a partial order relation on any set of non-zero positive integers }

$A = \{1, 3, 5\}$

$\div = \{(1,1), (1,3), (1,5)$
$(3,1), (3,3), (3,5)$
$(5,1), (5,3), (5,5)\}$

eg: Let 'C' be any collection of sets and "$\subseteq$" is set containment opⁿ (i.e. Subset opⁿ) then $(C, \subseteq)$ is a POSET.

$$C = \{ \underset{P}{\{\}}, \underset{Q}{\{1,2\}}, \underset{R}{\{1,2,3\}}, \underset{S}{\{2,3\}} \}$$

$$\subseteq = \{ (P,P), (P,Q), (P,R), (P,S), \cancel{(Q,P)}, (Q,Q), (Q,R), \cancel{(Q,S)}, \cancel{(R,P)}, \cancel{(R,Q)}, (R,R), \cancel{(R,S)}, \cancel{(S,P)}, \cancel{(S,Q)}, (S,R), (S,S) \}$$



Topic : Comparability

Let (A, R) be a POSET, {i.e. R is a Partial order rel^n}

For any pair of elements $a, b \in A$.

"a & b" are said to be comparable w.r.t. Partial order relation R if $a \, R \, b$ or $b \, R \, a$

i.e. $(a, b) \in R$ or $(b, a) \in R$

Slide

eg: Let $A = \{1, 2, 3, 4\}$

and $(A, \underset{R}{\div})$ is a POSET

1 divides every element of the set, i.e. '1' is Comparable with all elements of this set.

$(1,1) \in R$ ∵ '1 & 1' are Comparable

$(1,2) \in R$ ∵ 1 & 2 are —"—

$(1,3) \in R$, ∵ 1 & 3 — "—

$(1,4) \in R$, ∵ 1 & 4 —"—

$(2,1) \notin R$ but $(1,2) \in R$ ∵ "2 & 1" are Comparable

$(2,2) \in R$ ∵ '2 & 2' are Comparable

$(2,4) \in R$ ∵ "2 & 4" are Comparable

Neither 2 divides 3

nor 3 divides 2

i.e neither $(2,3) \in R$ nor $(3,2) \in R$

∴ "2 & 3" are not Comparable w.r.t. relation '$\div$' (divides)

Similarly "3 & 4" are also not Comparable w.r.t. relation divides ($\div$)



Topic : Totally Ordered Set

(Linearly ordered Set)

A POSET (A, R) is called a totally ordered set only if each pair of element of set A is comparable w.r.t. relation R. {and relation R is called Total order relation}

eg: let $A = \{1, 2, 4, 8\}$
$(A, \div)$ is a POSET
Every pair of elements of set A is Comparable w.r.t. relation $\div$ (divides)
∴ above POSET $(A, \div)$ is also a Totally ordered Set (TOSET)

eg: let $A = \{1, 2, 3, 4\}$
and $(A, \div)$ is a POSET.
We know '2 & 3' are not Comparable because neither 2 divides 3 (i.e. $(2,3) \notin \div$) nor 3 divides 2 (i.e. $(3,2) \notin \div$)
∴ $(A, \div)$ is just a POSET
it is not a Totally Ordered Set (TOSET)

Slide

eg: POSET $(A, \leq)$ is a TOSET for any set A of real numbers

eg: POSET $(A, \div)$ where A is any set of non-zero positive integers may or may not be a TOSET

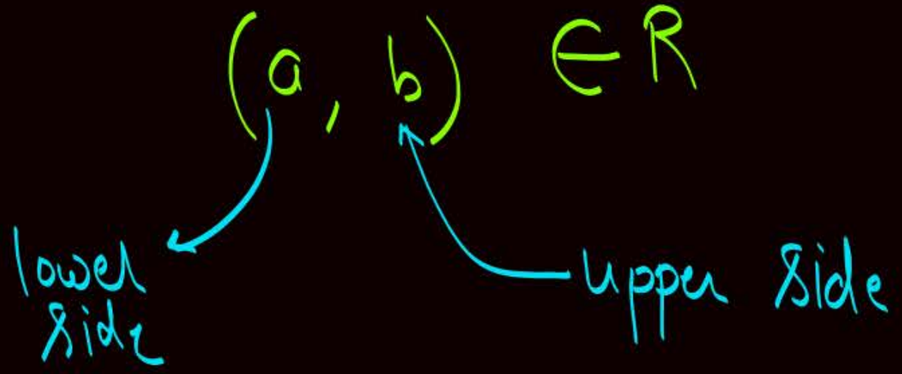
eg: if $A = \{1, 2, 4, 8\}$ then $(A, \div)$ is a TOSET

& if: $A = \{1, 2, 3, 4\}$ then $(A, \div)$ is not a TOSET

Note: Every TOSET is a POSET,
but every POSET need not be a TOSET.

- If $a \, R \, b$ i.e. $(a, b) \in R$

then a is said to be at lower side
& b is said to be at upper side





Topic : Least Upper Bound

/lub/ Join / Supremum

L.U.B.

it can be any partial order Rel^n

Let $(A,\leq)$ be a POSET, for any two elements $a,b\in A$ if there exists an element $c\in A$ such that,

$a\leq c$ and $b\leq c$, then c is called upper bound of a and b,

'c' is at the upper side of both a & b

And if there exists no element $d\in A$ such that

$a\leq d$ and $b\leq d$ and $d\leq c$,

then c is called least upper bound of a and b

There exists no element 'd' $\in A$ s.t. $\Rightarrow$ d is also an upper bound of a & b

and

d is at lower side of c

Slide



Topic : Least Upper Bound

lub / Join / Supremum

L.U.B.

Let (A,R) be a POSET, for any two elements $a,b \in A$ if there exists an element $c \in A$ such that,

$a \mathrel{R} c$ and $b \mathrel{R} c$, then c is called upper bound of a and b,

i.e. $(a,c) \in R$ & $(b,c) \in R$

And if there exists no element $d \in A$ such that

$a \mathrel{R} d$ and $b \mathrel{R} d$ and $d \mathrel{R} c$,

i.e. $(a,d) \in R$ & $(b,d) \in R$ & $(d,c) \in R$ then c is called least upper bound of a and b

Slide

Least upper bound of elements a & b is denoted by lub(a, b) or a ∨ b

lub(a, b) = lub(b, a)

∨ : Symbol used to denoted lub

a ∨ b = b ∨ a



Topic : Greatest Lower Bound

/glb/ Meet / Infimum

Let $(A, \leq)$ be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$c \leq a$ and $c \leq b$, then c is called lower bound of a and b,

'c' is at lower side of 'a' as well as 'b'

And if there exists no element $d \in A$ such that

$d \leq a$ and $d \leq b$ and $c \leq d$,

no element 'd' s.t. | d is at lower side of a & b | And | d is at upper side of c

then c is called greatest lower bound of a and b

Slide



Topic : Greatest Lower Bound

/glb/ Meet/ Infimum

Let $(A, \preceq)$ be a POSET, for any two elements $a,b \in A$ if there exists an element $c \in A$ such that,

$c \preceq a$ and $c \preceq b$, then c is called lower bound of a and b,

$(c,a) \in R$ & $(c,b) \in R$

And if there exists no element $d \in A$ such that

$d \preceq a$ and $d \preceq b$ and $c \preceq d$,

$(d,a) \in R$ & $(d,b) \in R$ & $(c,d) \in R$ then c is called greatest lower bound of a and b

Slide

Greatest lower bound of a & b is denoted by glb(a,b) or

|||

glb(b,a)

$a \wedge b$

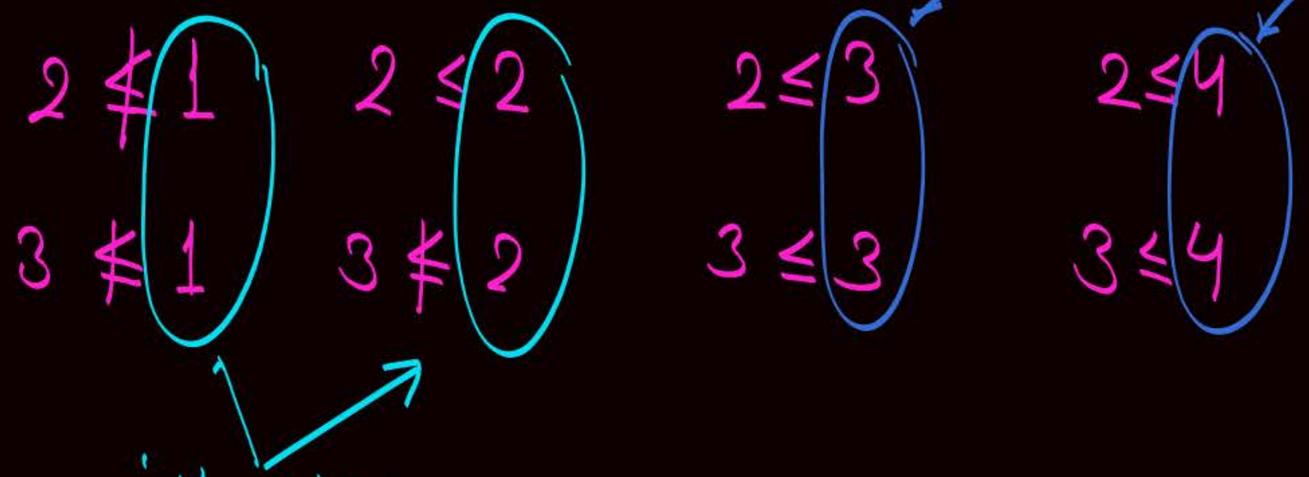
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$b \wedge a$

$\wedge$ → symbol to represent glb

eg: let $A = \{1, 2, 3, 4\}$
and $(A, \leq)$ is a POSET.

find lub & glb of elements $2, 3 \in A$



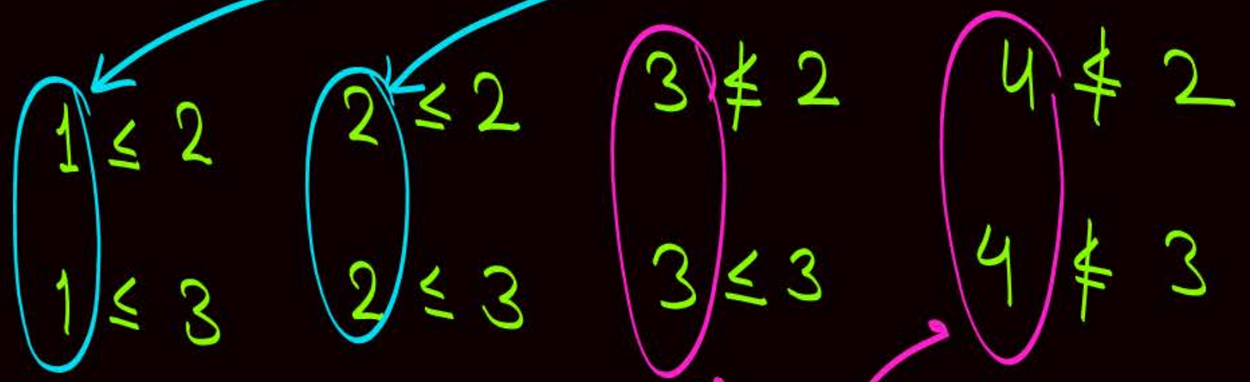
'1' & '2' can never be the upper bound of 2 & 3

Both 3 & 4 are upper bounds of 2 & 3

but $3 \leq 4$
i.e. 3 is at lower side of 4
∴ 4 can not be the least upper bound
∴ $lub(2,3) = 3$

eg: let $A = \{1, 2, 3, 4\}$
and $(A, \leq)$ is a POSET.

find lub & glb of elements $2, 3 \in A$



Both 1 & 2 are lower bounds of 2 & 3

but $1 \leq 2$

i.e. '1' is at lower side of another lower bound '2'

∴ 1 Can not be the greatest lower bound of 2 & 3

∴ glb(2,3) = 2

∴ 3 & 4 Can not be the lower bounds of 2 & 3

Note: * Let A is any set of real numbers and $(A, \leq)$ is a POSET,

$\leq$: less than or equal

for element $a, b \in A$

identify

(i) $lub(a,b) = ? \Rightarrow lub(a,b) = Max(a,b)$

(ii) $glb(a,b) = ? \Rightarrow glb(a,b) = Min(a,b)$

Note:

Let is a set of all natural numbers and $(A, \div)$ is a POSET.

for elements $a, b \in A$

(i) $lub(a, b) = LCM(a, b)$

(ii) $glb(a, b) = GCD(a, b)$

it is the set of all natural no.s

∴ LCM as well as GCD of any pair of natural No.s will also be present in the set of all natural numbers.

Q. Let $A = \{2, 3, 4\}$ and $(A, \div)$ is a POSET

What will be

lub (2,3) : No element in the set which is divided by 2 as well as by 3

∴ lub (2,3) does not exists in the above POSET

glb (2,3) = No element in the set which divides element '2' as well as element '3'

∴ glb (2,3) does not exists in the above POSET

Note: Lub and/or glb for a pair of elements of the set can not be out side the set

Q: Let A is any set of Non-zero positive integers and $(A, \div)$ is a POSET, than for elements $a, b \in A$,

$lub(a,b)$ = We don't know the elements of the set
∴ Nothing can be said

$glb(a,b)$ = We don't know the elements of the set
∴ Nothing can be said

Note:- Lub and/or Glb for a pair of elements of set 'A' can not be out side set A.

eg. Let $A = \{2, 3, 12, 24, 28\}$
and $(A, \div)$ is a POSET.

① lub (2, 3) = 12

② glb (2, 3) = does not exist

③ lub (24, 28) = does not exist

④ lub (3, 28) = does not exist

⑤ glb (24, 28) = 2

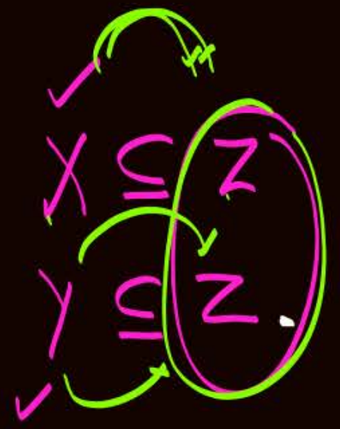
⑥ glb (3, 28) = does not exist

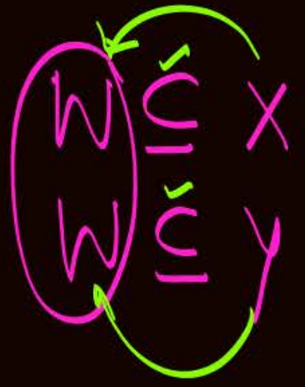
⑦ lub (12, 24) = 24

⑧ glb (12, 24) = 12

Note: Let A is any finite set, and P(A) is Power set of set A; then $(P(A), \subseteq)$ is a POSET.

P(A) — it is a collection of sets





Let X & Y are two elements of P(A) {i.e. $X, Y \in P(A)$}

then

$$lub(X, Y) = X \cup Y$$

$$glb(X, Y) = X \cap Y$$

let $A = \{1, 2, 3\}$

$P(A) = \{\{\}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

$(P(A), \subseteq)$ is a POSET.

$lub(\{1\}, \{2,3\}) = \{1,2,3\}$

$glb(\{1\}, \{2,3\}) = \{\} = \emptyset$

Note:- Let 'S' is any collection of sets, then $(S, \subseteq)$ is a POSET.

for any two sets $X, Y \in S$

$lub(X, Y) =$

$glb(X, Y) =$

} Can not be determined until we know the set 'S'.



$S = \{ \{1\}, \{a,b\}, \{1,a\}, \{1,2,a,b,c\} \}$

w.r.t POSET $(S, \subseteq)$

$\text{lub} \{\{1\}, \{a,b\}\} = \{1,2,a,b,c\}$ ✓

- $S = \{ \{1\}, \{a,b\}, \{1,a\} \}$

w.r.t. POSET $(S, \subseteq)$

$\text{lub}(\{1\}, \{a,b\}) =$ does not exist



Topic : Minimal elements in a POSET

If there is no element at lower side of any element then that element is called minimal element

$a \overset{R}{} b$ i.e. $(a, b) \in R$

lower side — upper side

Let $(A, \overset{R}{\leq})$ be a POSET

An element $x \in A$ is called a minimal element of POSET$(A, \leq)$ if there exists no element $y \in A$ other than x itself, such that $y \overset{R}{\leq} x$.

$(y, x) \in R$

lower

eg. let $A = \{2, 3, 4, 6, 15\}$

& $(A, \div)$ is a POSET.,

find minimal elements in above POSET.

2 & 3 are the minimal elements in above POSET

$(2, 4) \in \div$ ∴ 4 is not minimal

$(2, 6) \in \div$ ∴ 6 - ,, ——

$(3, 15) \in \div$ ∴ 15 - ,, ——

eg: let $A = \{1, 2, 3, 4, 6, 12\}$

& $(A, \div)$ is a POSET

find minimal elements in above POSET.

'1' is the only minimal element in above POSET.



Topic : Maximal elements in a POSET

If there is no element at upper side of an element, then that element is called maximal element

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a maximal element of POSET$(A, \leq)$ if there exists no element $y \in A$ other than x itself, such that $x \leq y$.

$x \; R \; y$

upper

Slide

eg. let $A = \{2, 3, 4, 6, 15\}$

& $(A, \div)$ is a POSET.,

find maximal elements in above POSET.

- 4, 6 & 15 are maximal elements in above POSET.

eg: let $A = \{1, 2, 3, 4, 6, 12\}$

& $(A, \div)$ is a POSET

find maximal elements in above POSET.

- '12' is the only maximal element in above POSET.



Topic : Minimum element of a POSET / (Least element)

Let $(A, \leq)$ be a POSET

An element $m \in A$ is called a minimum element of POSET$(A, \leq)$

if and only if

$m \leq a, \forall a \in A$

An element of the POSET, which is at lower side of all other elements of the POSET is called minimum element of POSET

i.e. $(m, a) \in \leq_R, \forall a \in A$

lower side

eg. let $A = \{2, 3, 4, 6, 15\}$

& $(A, \div)$ is a POSET.

find minimum element of above POSET. {if exist}

. There is no element which divides all elements of above set

∴ Minimum element does not exist in above POSET.

eg: let $A = \{1, 2, 3, 4, 6, 12\}$

& $(A, \div)$ is a POSET

find minimum element of above POSET. {if exist}

'1' is the minimum element of above POSET.



Topic : Maximum element of a POSET

↳ An element of the POSET, which is at upper side of every other element of the POSET is called maximum element of POSET

Let $(A, \leq)$ be a POSET

An element $n \in A$ is called a maximum element of POSET$(A, \leq)$

if and only if

$a \leq n, \forall a \in A$

$(a, n) \in$ Relation $(\leq)$, $\forall a \in A$

Slide

eg. let $A = \{2, 3, 4, 6, 15\}$

& $(A, \div)$ is a POSET.,

find maximum element of above POSET. {if exist}

Maximum element does not exist in the above POSET.

eg: let $A = \{1, 2, 3, 4, 6, 12\}$

& $(A, \div)$ is a POSET

find maximum element of above POSET. {if exist}

'12' is the maximum element of above POSET.



Topic : NOTE

① There exists at least one / One or more minimal elements in a POSET {w.r.t finite set}

Similarly, ② There exists at least one / one or more maximal elements in a POSET {w.r.t. finite set}

③ There exist at most one / 0 or 1 minimum element in a POSET.

④ There exist at most one / 0 or 1 maximum element in a POSET.

Slide



Topic : NOTE

⑤ If there exists two or more minimal elements in a POSET, then minimum element does not exist in that POSET

⑥ If there exists two or more maximal elements in a POSET, then maximum element does not exist in that POSET.

Slide



Topic : NOTE

(7) Every minimum element is also a minimal element of POSET, but converse of the statement need not be true.

(8) Every maximum element is also a maximal element of POSET, but converse of the statement need not be true

Slide



Topic : NOTE

(9) If there exists a unique minimal element in a POSET, then it is also the minimum element of POSET.

(10) If there exists a unique maximal element in a POSET, then it is also the maximum element of POSET

Slide

Note:- In a POSET lub and/or glb need not exist for every pair of elements of the set



Topic : Join Semi Lattice

lub

A POSET in which Least upper bound (lub) exists for every pair of elements is called a "Join" semi lattice.

{"glb" may or may not exist}

eg: let $A = \{2, 3, 4, 6, 12\}$ & $(A, \div)$ is a POSET.

lub exist for every pair of elements of POSET $(A, \div)$

∴ it is a Join semi lattice

Slide

eg: $A = \{2, 3, 4, 6, 15\}$ & $(A, \div)$ is a POSET

"lub" does not exist for many pairs of elements. {eg. lub (3, 4)}

∴ it is not a join semi lattice

eg: $A = \{1, 2, 3, 4, 6, 12\}$ & $(A, \div)$ is a POSET.

'lub' exists for every pair of elements of the set

∴ it is a Join semi lattice



glb

Topic : Meet Semi Lattice

A POSET in which Greatest lower bound (Glb) exist for every Pair of elements is called a Meet Semi lattice {lub may or may not exist}

eg: $A = \{1, 2, 3, 4, 6\}$ & $(A, \div)$ is a POSET.

glb exist for every pair of elements of POSET

∴ It is a Meet Semi lattice

Slide

① $A = \{2, 3, 4, 6, 12\}$
$(A, \div)$ is POSET

→ lub exist for every pair of elements ∴ Join semi lattice
→ glb does not exist for some pair of elements ∴ Not a Meet Semi lattice

② $B = \{1, 2, 3, 4, 6\}$
$(B, \div)$ is POSET

→ lub does not exist for some pair ∴ Not a Join semi lattice
→ glb exist for every pair of elements ∴ Meet semi lattice

③ $C = \{2, 3, 4, 6, 15\}$
$(C, \div)$ is POSET

→ lub does not exist for some pair of elements ∴ Not a Join Semi lattice
→ glb does not exist for some pair of elements ∴ Not a Meet semi lattice

④ $D = \{1, 2, 3, 4, 6, 12\}$
$(D, \div)$ is POSET

→ lub exist for every pair of elements ∴ Join semi lattice
→ glb exist for every pair of elements ∴ Meet semi lattice



Topic : Lattice

→ A POSET which is both Join semi lattice as well as a Meet semi lattice is called a lattice

(Or)

. A POSET in which both lub as well as glb exist for every pair of elements is call a lattice

Slide

① $A = \{2, 3, 4, 6, 12\}$
$(A, \div)$ is POSET

- lub exist for every pair of elements ∴ Join semi lattice
- glb does not exist for some pair of elements ∴ Not a Meet Semi lattice

] Not a lattice

② $B = \{1, 2, 3, 4, 6\}$
$(B, \div)$ is POSET

- lub does not exist for some pair ∴ Not a Join semi lattice
- glb exist for every pair of elements ∴ Meet semi lattice

] Not a lattice

③ $C = \{2, 3, 4, 6, 15\}$
$(C, \div)$ is POSET

- lub does not exist for some pair of elements ∴ Not a Join Semi lattice
- glb does not exist for some pair of elements ∴ Not a Meet semi lattice

] Not a lattice

④ $D = \{1, 2, 3, 4, 6, 12\}$
$(D, \div)$ is POSET

- lub exist for every pair of elements ∴ Join semi lattice
- glb exist for every pair of elements ∴ Meet semi lattice

] It is a lattice



2 mins Summary

- **Topic** Minimal / Maximal elements in a POSET
- **Topic** Minimum(Least) element of POSET
- **Topic** Maximum(Greatest) element of POSET
- **Topic** Lattice
- **Topic** Types of lattice will be discussed along with Hasse diagram.

THANK - YOU